Lemma 1.45 (Right Angles on Incircle Chord)

Kagebaka
3001 posts

Oct 4, 2018, 6:46 PM • 5 #1

The incircle of ABC is tangent to $\overline{BC}, \overline{CA}, \overline{AB}$ at D, E, F, respectively. Let M and N be the midpoints of $\overline{BC}$ and $\overline{AC}$, respectively. Ray BI meets line EF at K. Show that $\overline{BK} \perp \overline{CK}$. Then show K lies on line MN.

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Oct 4, 2018, 6:55 PM • 3 #2

Solution Part 1

Let $\angle ABC = 2x$. Since $\triangle CMN \sim \triangle CBA$ by SAS, $MN \parallel AB$, and drawing the angle bisector of $\angle ABC$, we find that $\angle BKM = x$ by alternating interior angles. However, this implies that $\angle KBC = \angle BKM = x \implies KM = BM$, so BKC must be a right triangle because the median length is half of the hypotenuse length.

I wasn't able to prove that K lies on MN, however. My approach was to prove that AFE is isosceles by proving that $\angle FKB = \dfrac{\angle ACB}{2}$, but so far it hasn't been working :(

This post has been edited 2 times. Last edited by Kagebaka, Oct 4, 2018, 6:59 PM

Arthur.
133 posts

Oct 7, 2018, 5:34 PM • 2 #3

> *Kagebaka wrote:*
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Doesn't this assume K, M and N are collinear?

Kagebaka
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Oct 7, 2018, 8:02 PM • 2 #4

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